Servo Characterization for the SOFIA Modular Kinetic Surface

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***Abstract*—**

***Index Terms*—**

I. Introduction

SOFIA is a modular robotic installation composed of 16 mechatronic units arranged in a two-dimensional 4×4 array, with each unit providing two rotational degrees of freedom actuated by servomotors. The system aims to achieve coordinated control of the multiple actuators while enabling interaction through static and dynamic human gestures. For this purpose, SOFIA integrates embedded control, inter-module communication, and an external computer vision system for gesture recognition.

The SOFIA system integrates three main capabilities: automatic configuration and communication of modules through a Modbus network, characterization of the mechanical time constant of the servomotors, and recognition of static and dynamic human gestures through computer vision. These capabilities require the integration of mechanical actuation, servo control, embedded processing, and communication subsystems. Since each of the 16 units uses two servomotors, the complete demonstrator requires the coordinated control of 32 actuators and the generation of multiple PWM signals. The embedded controller must coordinate the desired actuator positions, while the communication architecture should support the future integration of additional SOFIA modules without significantly increasing wiring and control complexity.

Motorized arrays have been widely explored as a mechanism for generating dynamic physical surfaces. Relief introduced an actuated tabletop composed of 120 individually addressable motorized pins and emphasized a scalable hardware and software architecture [1]. This concept was extended through inFORM, where shape actuation was employed to generate dynamic physical affordances and constraints and to manipulate physical objects [2]. At a larger scale, TRANSFORM explored the transformation of static physical structures into dynamic systems driven by streams of data and energy [3]. These approaches demonstrate the potential of coordinated arrays of actuators, although their architectures primarily focus on large pin-based or installation-scale physical surfaces.

Alternative approaches have explored different actuation principles and modular configurations. PneUI demonstrated shape-changing interfaces based on pneumatically actuated soft composite materials, integrating sensing and active shape output within deformable structures [4]. In contrast, LineFORM investigated electromechanical shape transformation through serial chains of one-degree-of-freedom servo actuators [5]. This concept evolved toward greater hardware modularity in ChainFORM, where identical modules incorporate motor actuation, sensing, visual output, and embedded processing [6]. These systems highlight modularity and distributed actuation as relevant design characteristics for reconfigurable physical interfaces.

The main technological characteristics and differences identified among the reviewed shape-changing interfaces are summarized in Table I.

In contrast to the reviewed shape-changing interfaces, SOFIA combines a two-dimensional 4×4 arrangement with two rotational degrees of freedom per unit. Unlike pin-based displays, which predominantly constrain motion to a single translational axis, or line-based modular systems organized as serial chains, each SOFIA unit can independently modify its orientation while remaining part of the same physical surface. This configuration introduces design challenges related to coordinated actuation, modularity, scalability, control architecture, electrical integration, and communication. Based on the reviewed literature, these characteristics were identified as relevant criteria for evaluating the technological architecture of the demonstrator. Table I summarizes the comparative analysis of the reviewed systems, while Table II contrasts SOFIA with the reviewed solutions using the established comparison criteria.

From the functional decomposition of the demonstrator and its comparison with the literature, several technological opportunities were identified. One is the development of a scalable control architecture capable of independently coordinating all actuators while reducing the processing load on the main controller. Another is the implementation of an inter-module communication architecture that allows multiple SOFIA modules to operate as part of a larger system. Additional opportunities include the implementation of software-defined motion limits and smoothing strategies to prevent abrupt servo movements and reduce mechanical stress.

Based on these opportunities, this work focuses on defining and justifying a preliminary electronic and control architecture for the SOFIA demonstrator. The main areas considered are actuator control, PWM generation, embedded processing, inter-module communication, electrical compatibility, and scalability. These criteria provide the basis for the selection and experimental evaluation of the technologies used in the subsequent development stages of the SOFIA system.

II. Materials and Methods

The SOFIA demonstrator consists of sixteen electromechanical units, each with two rotational degrees of freedom. The reference architecture requires thirty-two servo channels

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TABLE I
COMPARATIVE MATRIX OF RELATED SHAPE-CHANGING INTERFACE SYSTEMS

Reference	Method / technology	Metric / focus	Main result	Gap with respect to SOFIA
Relief [1]	Array of 120 motorized pins, scalable low-cost control	Actuation density, hardware/software scalability	Continuous surface capable of rendering and animating 3D shapes, with per-pin input sensing	Translational actuation only (1 DOF per pin); no independent rotation per unit
inFORM [2]	Pin-based shape display on a pre-existing platform	Dynamic physical affordances and constraints, object manipulation	Real-time direct-touch interaction and physical rendering of content	Same translational limitation as Relief; centralized architecture, not modular per unit
TRANSFORM [3]	Three large-scale pin displays, kinetic energy as input	Aesthetic transformation from static furniture to dynamic system	Over one thousand pins coordinated in real-time wave motion	Installation-scale system; no per-module orientable control or distributed communication between units
PneUI [4]	Pneumatically actuated multi-layer soft composite materials	Integrated input sensing and shape output in the material itself	Four functional applications (phicons, mobile, case, lamp) with controllable shape change	Soft/pneumatic actuation, not discrete rotational motion; difficult to scale to precise angular control
LineFORM [5]	Serial chain of one-DOF servo actuators	Line/curve-based interaction affordances	Cord, mobile, and body-constraint interfaces through line-shaped actuation	Serial (chain) configuration, not a two-dimensional array; 1 DOF per unit, not 2
ChainFORM [6]	Identical modules with servo, touch sensing, visual output and embedded processing	Hardware modularity and reconfigurability	Modular, reconfigurable platform with rich per-module I/O	Modularity demonstrated, but still a linear chain, not a 2D array with 2 DOF per node

TABLE II
BENCHMARKING AGAINST RELATED SHAPE-CHANGING SYSTEMS

Solution	Configuration	DOF/ unit	Modularity / scalability
inFORM [2]	Pin array on fixed platform	1 (translational)	Low – centralized control, not designed for modular expansion
LineFORM / ChainFORM [5], [6]	Serial chain of modules	1 per module	Medium – modular, but limited to a linear configuration
PneUI [4]	Pneumatic soft composite material	Continuous (non-discrete)	Low – difficult to discretize into individually addressable nodes
SOFIA (proposed)	2D array, 4×4 modules	**2 (rotational)**	**High – I2C bus + RS-485 designed for future module expansion**

TABLE III
SOFIA MAIN REQUIREMENTS

Requirement	Specification	Verification
Power	5 V servo supply	Compatible with MG90S range
Logic voltage	3.3 V MCU logic	Required for ESP32-C3 GPIO
Communication	WiFi/Bluetooth + I2C	Supported by ESP32-C3
PWM control	32 servo channels, 50 Hz	Two PCA9685 drivers required
Mechanical	16 modules, 2 DOF each	32 servos required

controlled by one ESP32-C3 and two PCA9685 PWM drivers [7].

The objective of this document is to verify the compatibility of the selected components according to electrical, communication and mechanical requirements. The verification criteria are based on the hardware requirements and checklist defined in the technical session [8]. The document structure follows the LabDST documentation requirements for technical datasheets and component verification reports [9].

The main electrical, communication, and mechanical requirements of the SOFIA demonstrator are summarized in Table III.

TABLE IV
ESP32-C3 MAIN SPECIFICATIONS

Parameter	Value	Verification
Supply voltage	3.0–3.6 V	Requires regulated 3.3 V supply
WiFi	802.11 b/g/n 2.4 GHz	Complies
Bluetooth	BLE 5	Complies
Logic level	3.3 V GPIO	5 V cannot be applied directly
Boot pins	GPIO2, GPIO8, GPIO9	PCB design verification required
Temperature	Up to 105 °C	Suitable for laboratory operation

TABLE V
PCA9685 MAIN SPECIFICATIONS

Parameter	Value	Verification
Channels	16 PWM channels	Two devices provide 32 outputs
Resolution	12 bits	Complies
Interface	I2C Fast Mode Plus	Compatible with ESP32-C3
Voltage	2.3–5.5 V	Use 3.3 V logic
Frequency	24–1526 Hz	Configure to 50 Hz for servos
Addressing	Up to 62 devices	Use different addresses (0x40/0x41)

A. ESP32-C3 Technical Analysis

The main technical specifications of the ESP32-C3 microcontroller are presented in Table IV.

Source: ESP32-C3 Datasheet, Espressif Systems. [10]

B. PCA9685 Technical Analysis

The specifications of the PCA9685 PWM driver are summarized in Table V.

Source: PCA9685 Datasheet, NXP. [11]

C. MG90S Servo Technical Analysis

The nominal operating characteristics of the MG90S servomotor are presented in Table VI.

Source: TowerPro MG90S documentation. [12]

D. Checklist Verification

The preliminary component compatibility assessment is summarized in Table VII.

TABLE VI
MG90S MAIN SPECIFICATIONS

Parameter	Value	Verification
Voltage	4.8–6.0 V	Compatible with 5 V rail
Torque	1.8–2.2 kgf cm	validation required
Speed	0.08–0.10 s/60°	Reference value
Mass	13.4 g	Must be considered in CAD
PWM frequency	50 Hz	Compatible with PCA9685
Stall current	Not reported	Measurement required

TABLE VII
VERIFICATION SUMMARY

Component	Status	Comment
ESP32-C3	**Complies**	Requires 3.3 V regulation and boot verification
PCA9685	**Complies**	Requires unique I2C addresses and 50 Hz configuration
MG90S	**Partial**	Torque and current require validation

TABLE VIII
BILL OF MATERIALS SUMMARY

Item	**Component**	**Qty**	**Manufacturer**
1	Sofia003-P001-03	16	3D Printed
2	MG90S servo	32	Tower Pro
3	Sofia003-E002-01	16	3D Printed
3.1/4.4	Pro3	32	Tower Pro
3.2	Sofia003-P004-02	16	3D Printed
3.3	Sofia003-P004-03	16	3D Printed
3.4/3.5/ 4.3/7/8	M2 screws (various)	160	Walfront
3.6	Axis 1 9mm x 9mm	16	Mae
3.7	Sofia003-P002-02	16	3D Printed
4	Sofia003-E003-01	16	3D Printed
4.1	Sofia003-P005-01	16	3D Printed
4.2	Sofia003-P005-02	16	3D Printed
5	Sofia003-P008-01	16	3D Printed
6	Sofia003-P009-01	16	3D Printed
9	Inser M3x5x4	16	ZWMSSLL
10	Sofia003-P000-01	16	3D Printed
-	ESP32-C3 SuperMini	1	Espressif
-	PCA9685 Servo Driver	2	NXP
-	MAX13487E TTL485-V2.0	16	Analog Devices Inc.
-	LED RGB	1	Mexbit
-	JST-PH Male	1	JTAREA
-	JST-PH Female	1	JTAREA

E. Bill of Materials

The components required for the SOFIA demonstrator, including mechanical parts, actuators, and electronic devices, are summarized in Table VIII.

F. Justification of Technological Selection

As summarized in Table IX, the technological architecture was selected according to the functional and scalability requirements of the SOFIA demonstrator. MG90S servomotors were selected to provide controlled angular positioning for each degree of freedom, while two PCA9685 drivers provide the 32 PWM channels required by the system through a shared I2C bus. The ESP32-C3 SuperMini was selected as the main controller due to its integrated wireless connectivity and support for the required communication interfaces. Finally, RS-485 was selected for inter-module communication because its differential and multi-node architecture supports the future connection of multiple SOFIA modules.

TABLE IX
COMPARATIVE MATRIX OF ALTERNATIVES

Function	Alternatives considered	Selected	Selection criterion
Actuation	Servo / Stepper / DC	MG90S	Integrated angular positioning
PWM generation	MCU PWM / PCA9685 / Additional MCU	PCA9685	16 PWM channels through I2C
Processing	Arduino / STM32 / ESP32-C3 SuperMini	ESP32-C3 SuperMini	Wireless connectivity + required interfaces
Inter-module communication	UART / CAN/ RS-485	RS-485	Differential multi-node communication

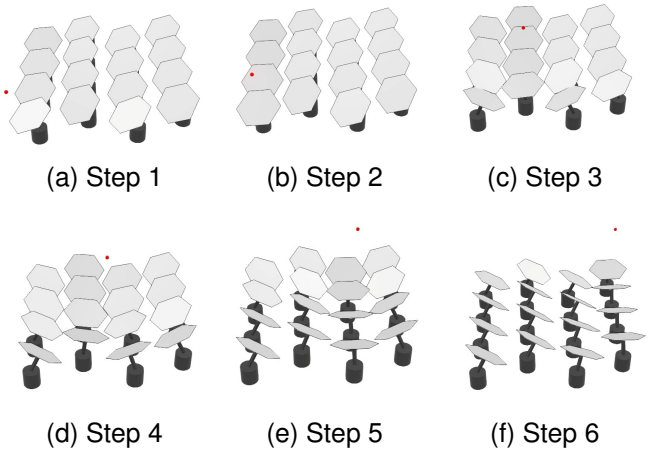


Fig. 1. Simulation of the SOFIA module tracking a moving reference point using inverse kinematics.

III. DESIGN AND IMPLEMENTATION OF THE TECHNOLOGICAL SOLUTION

Figure 1 shows the simulated response of the sixteen-module SOFIA structure while tracking the moving point. The sequence illustrates each module reorientation at different instants, where each module adjusts its two rotational degrees of freedom to maintain alignment with reference position.

Algorithm 1 summarizes the inverse kinematics and smoothing strategy used by each SOFIA module to orient its platform toward the moving target.

IV. RESULTS

V. DISCUSSIONS

VI. CONCLUSIONS

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Algorithm 1 SOFIA module target tracking control

Require: Target position p, module position r_0, joint limits $q_{1min}, q_{1max}, q_{2min}, q_{2max}$, smoothing gain K

Ensure: Servo angles q_1, q_2

$v \leftarrow p - r_0$
$\hat{v} \leftarrow (v/\|v\|)$
$q_1^* \leftarrow atan2(-\hat{v}_y, \hat{v}_z)$
$q_2^* \leftarrow asin(\hat{v}_x)$
$q_1^* \leftarrow Clamp(q_1^*, q_{1min}, q_{1max})$
$q_2^* \leftarrow Clamp(q_2^*, q_{2min}, q_{2max})$
$q_1 \leftarrow q_{1old} + K(q_1^* - q_{1old})$
$q_2 \leftarrow q_{2old} + K(q_2^* - q_{2old})$
Apply q_1, q_2 to servo motors
return q_1, q_2

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